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10 Questions

Ready to review what you've learned before starting the assignment? We're here to help.

1. When is the normal approximation for $\frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$ most accurate? (1 point)

☐ When the posterior distribution is unimodal.

☐ When the sample size is small.

☒ When the posterior distribution is unimodal and symmetric.

☐ When the posterior distribution is skewed and asymmetric.

Incorrect

Correct. The normal approximation is most accurate when the posterior distribution is unimodal and symmetric, and the sample size is large.

2. What does asymptotic theory focus on in large sample properties?

☐ Properties of Bayesian methods for small sample sizes.

☐ Prior distributions for small samples.

☐ Prior posterior distributions.

☒ Behavior of posterior distributions as the sample size goes to infinity.

Correct

Correct. Asymptotic theory focuses on large sample properties.

3. Which of the following is the most compelling reason to use a normal approximation of a distribution?

☐ It ensures the posterior distribution is always unimodal.

☐ It is the only method available for large datasets.

☐ It always provides the most accurate results.

☒ It simplifies the mathematical computations involved.

Correct

Correct. One of the most compelling reasons to use a normal approximation is that it simplifies the mathematical computations, making it easier to handle complex problems.

4. What does the Bayesian mode correspond with from the frequentist perspective?

☐ The maximum likelihood estimate (MLE).

☒ The maximum likelihood estimate (MLE).

☐ The mean of the sampling distribution.

☐ The median of the sampling distribution.

Correct

The Bayesian mode corresponds with the maximum likelihood estimate (MLE) from the frequentist perspective, especially when using a normal distribution prior.

5. What is the impact of increasing sample size on the influence of the prior distribution in Bayesian analysis?

☐ The influence of the prior increases.

☒ The influence of the prior decreases.

☐ The influence remains the same.

☐ The influence becomes negligible only if the prior is informative.

Correct

Correct. The posterior is a balance between the prior and the data. As the data grows larger, however, it dominates.

6. What is the primary difference between $\pi(\theta|y)$ and $\pi(\theta)$?

☐ $\pi(\theta)$ measures prior information, while $\pi(\theta|y)$ measures posterior information.

☐ $\pi(\theta)$ is used in Bayesian analysis, while $\pi(\theta|y)$ is used in frequentist analysis.

☐ $\pi(\theta)$ is always larger than $\pi(\theta|y)$ as the sample size goes to infinity.

☒ $\pi(\theta)$ is the observed information, while $\pi(\theta|y)$ is the expected information.

Correct

Correct. $\pi(\theta)$ refers to the observed information, which is calculated from the data, while $\pi(\theta|y)$ refers to the expected information, which is based on the expected value over the sampling distribution.

7. Why does Theorem 1 assume the parameter space is finite and not countably infinite?

☐ To ensure the prior distribution is proper.

☐ To simplify the mathematical computations.

☐ To ensure the posterior distribution is always normal.

☒ To avoid numerical integration issues.

Incorrect

While true, the theorem assumes the parameter space is finite and not countably infinite to ensure the posterior distribution is proper.

8. Why might a normal approximation be more accurate for marginal distributions than joint distributions?

☐ Marginal distributions are independent.

☒ Averaging over components can bring distributions closer to normality.

☐ Joint distributions have fewer components.

☐ Marginal distributions are always normal.

Correct

Correct. Averaging over components can bring distributions closer to normality, which is why normal approximations are often more accurate for marginal distributions.

9. What is the purpose of frequency evaluations of Bayesian inferences?

☒ To assess the performance of Bayesian methods under repeated sampling.

☐ To simplify Bayesian computations.

☐ To validate prior distributions.

☐ To determine the exact posterior distribution.

Correct

Correct. Frequency evaluations assess how well Bayesian methods perform when repeated samples are taken.

10. Why is $\pi(\theta)$ not a uniform prior for the normal distribution?

☐ It does not incorporate any prior knowledge.

☒ It does not provide a proper prior distribution.

☐ It leads to a biased posterior distribution.

☐ It ensures the posterior distribution is always uniform.

Correct

Correct. $\pi(\theta)$ is not a proper prior for the normal distribution because it does not integrate to a finite value over the parameter space, leading to an improper prior.

11. Why do we use log odds in the asymptotic proofs for Chapter 4?

☐ To convert probabilities into percentages.

☐ To improve the accuracy of small sample estimates.

☐ To ensure that all values are positive.

☒ To make the models linear.

Correct

Correct. Using log odds in asymptotic proofs helps make the models linear, which simplifies mathematical derivations and helps in handling the properties of the distributions.

12. Based on the proof done with normal approximation, is it possible to compute a Beta approximation?

☐ No, because normal approximations are always superior.

☐ Yes, if the data are transformed to fit a Beta distribution.

☒ No, because Beta approximations require different assumptions and parameters.

☐ Yes, because normal and Beta distributions are interchangeable.

Correct

Correct. Beta approximations are used for distributions bounded between 0 and 1 and involve different assumptions and parameters than normal approximations.

13. What is the primary use of normal approximations in Bayesian analysis?

☒ Simplifying calculations.

☐ Increasing sample size.

☐ Increasing precision.

☐ Ensuring consistency.

Correct

Correct. Normal approximations help simplify the computations involved in Bayesian analysis.

14. Why do we study the log posterior?

☒ To simplify the multiplication of probabilities.

☐ To ensure the posterior is a normal distribution.

☐ To make it easier to compute the posterior mean.

☐ To simplify the mathematical form and make derivation easier.

Incorrect

While true, the log posterior simplifies addition, not multiplication, of probabilities.

15. Why are heavy tails not an issue with normal approximation?

☐ Normal approximation adjusts for heavy tails through scaling.

☒ Normal approximation only considers the central part of the distribution.

☐ Heavy tails are smoothed out by the central limit theorem.

☐ Heavy tails do not affect the overall tendency of the distribution.

Correct

Correct. Normal approximation primarily focuses on the central part of the distribution where most of the data lies, and the influence of heavy tails is minimized in this region.